LOUIS BORJA

PORTFOLIO

Mechanical & Manufacturing Engineer

Industrial Systems | Modelling & Process Optimisation



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01 / Process Modelling — HALEON

Production Unit Modelling on AnyLogic

Built a simulation model of a full production unit in AnyLogic to assess how a new production line would affect logistics and material flows across the site.



Model at a glance



Key contributions

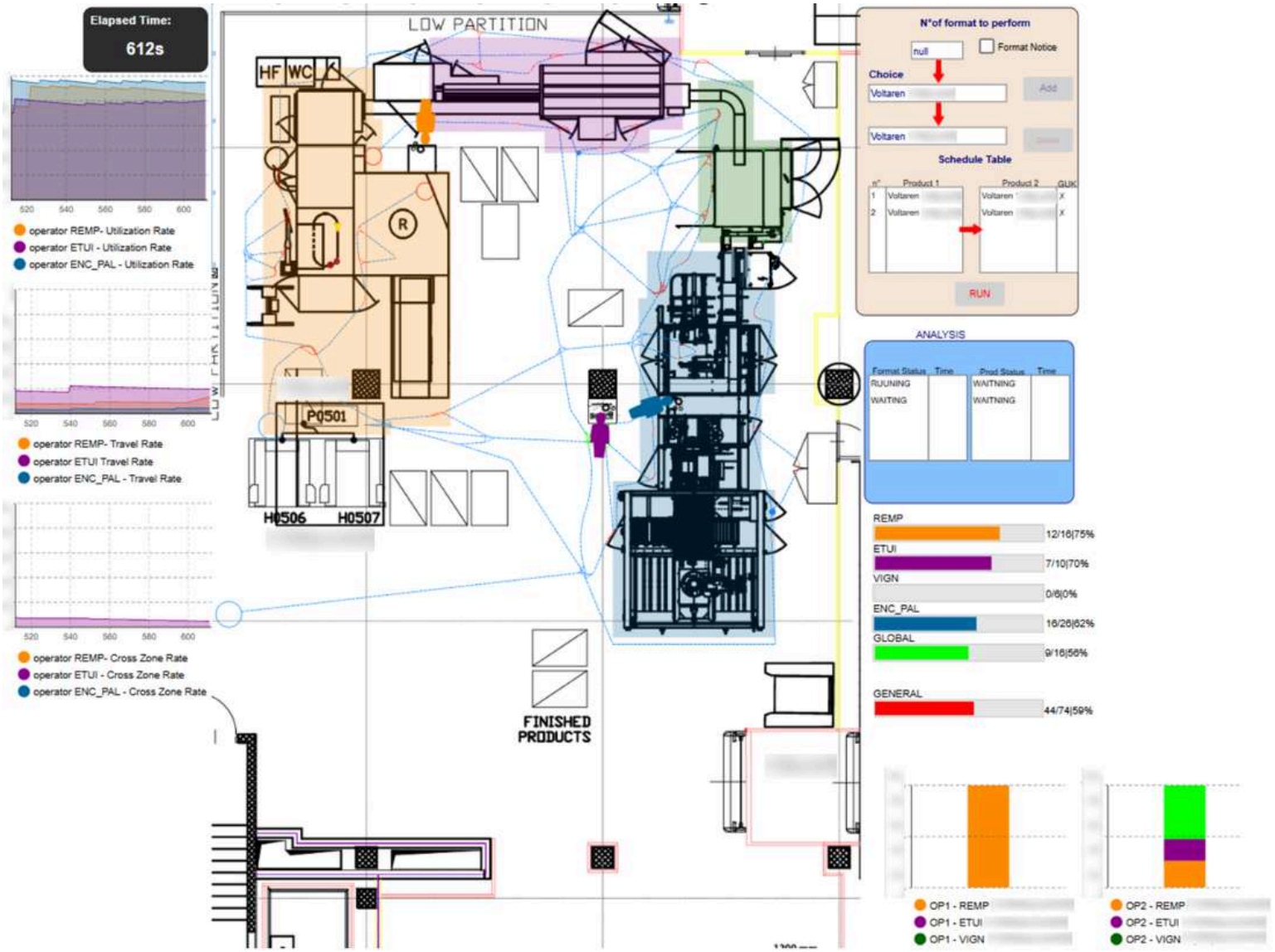
01 Modelled **material flows, transport routes and resource use** to identify bottlenecks and capacity limits.

02 Simulated **format changeovers** to define **standard operating procedures** and set realistic changeover times.

01 / Changeover Model Architecture

Single-line changeover optimisation

Built a changeover optimisation simulation to test multiple product scenarios and optimise the task sequence to minimise changeover time.



Model at a glance



Testable scenarios
multi-product mixes

Task sequence
optimisation

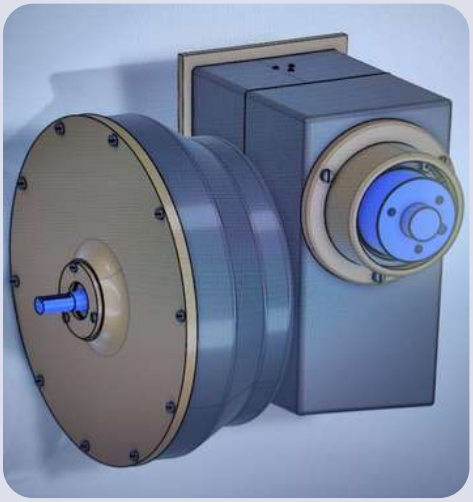
KPI export
to Excel

Reduced line
downtime

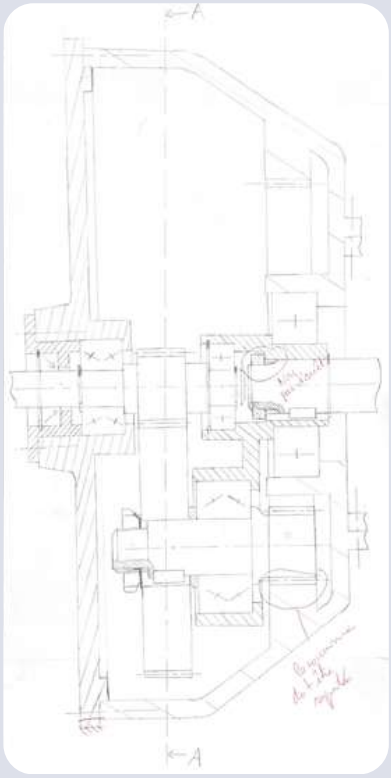
02 / Computer-Aided Design (CAD)

Design and CAD Modelling of a Planetary Gear Train

Designed a planetary gear reducer for a tribology test rig from hand sketch to a full CAD assembly, including kinematic validation, detailed drawings and assembly sequencing.

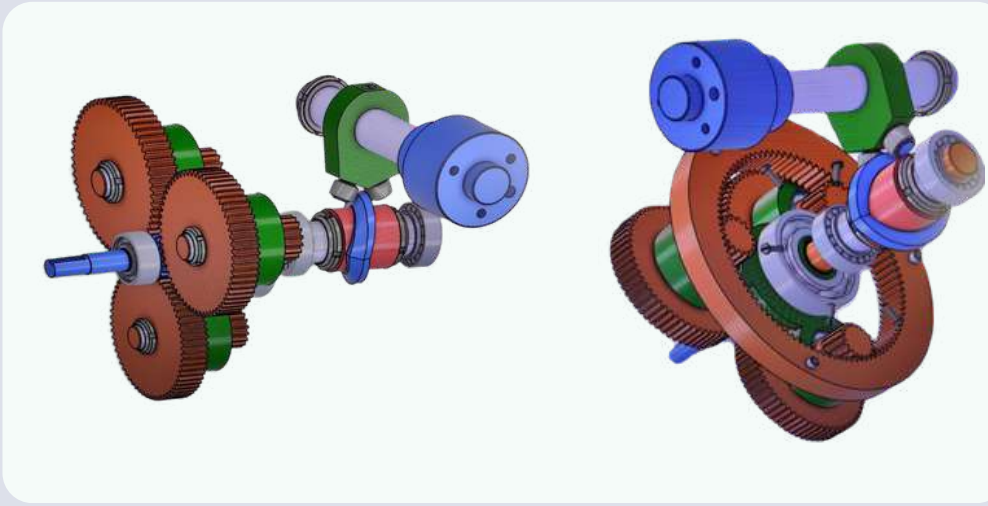


Final assembly with housing
3DEXPERIENCE CATIA



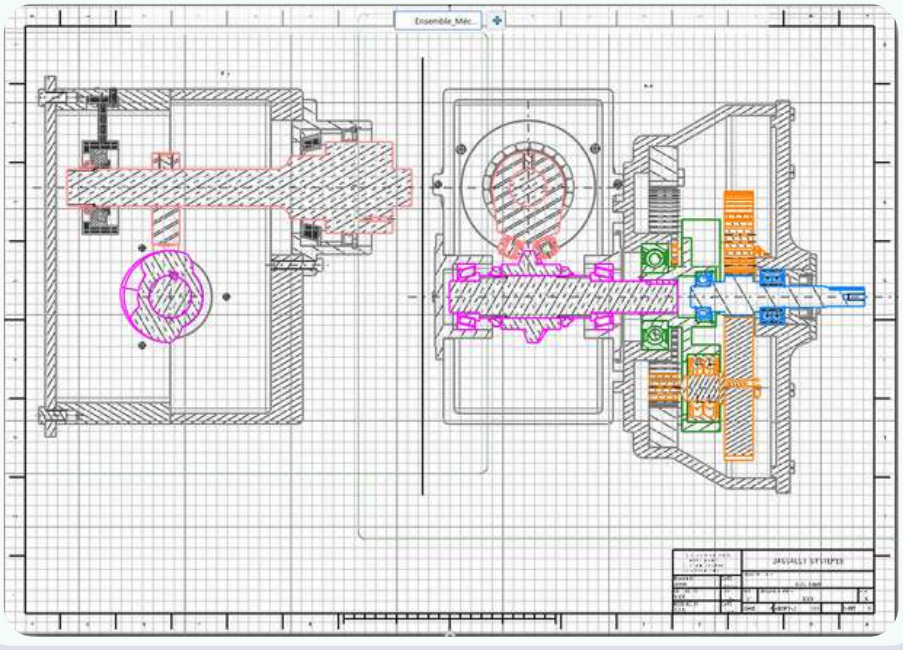
01 Concept

Hand technical drawing, section A-A



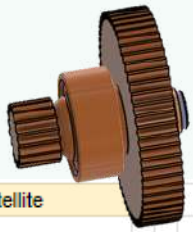
02 Kinematics

Sun, planet and ring gear train validated without housing



03 Detailed design

CAD technical drawing: sections, fits and part identification



Arbre satellite

Ordre de montage → Repérage des pièces ↓	1	2	3	4	5	6	7	8
14 - R3	x							
21 - entretoise satellite		x						
21 - clavette satellite			x					
16 - Z2				x				
25 - ecrou encoche sat					x			

04 Assembly routing

Assembly sequence for each subassembly (planet shaft)

03 / Machining — Single-Cylinder Crankshaft

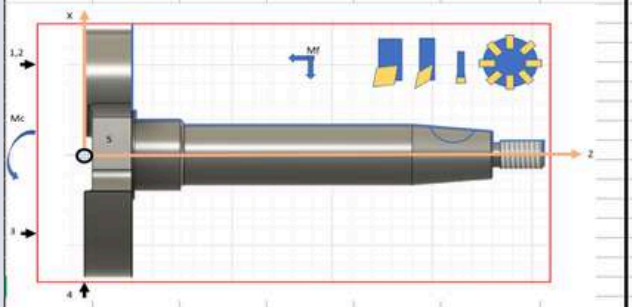
From process planning to a machined part

Planned and programmed the machining of a single-cylinder crankshaft on a CNC mill-turn centre: process plan with cutting parameters, workholding definition, CAM toolpaths in Fusion 360 and machining of the final part.



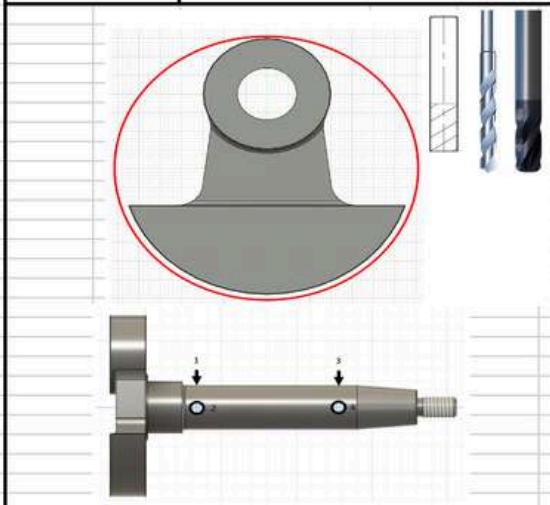
Machined crankshaft — CNC mill-turn

PHASE N°10 — Element : Vilebrequin — Machine : centre d'usinage millturn 5 axes



Opération :	Outil :	Vc	f	ap
dressage face droite	outil carbure εr=85	100	0,3	1
finition dressage face droite	outil carbure εr=35	150	0,1	0,5
ébauche chariotage/dressage	outil carbure εr=85	100	0,3	1
contournage finition	outil carbure εr=35	150	0,1	0,5
rainurage	outil à rainurer 2mm	120	0,1	2
filetage M8	outil à fileter	/	1,25 (pas M8)	
fraisage clavette	fraise 3 tailles R8 e3	120	0,05	/

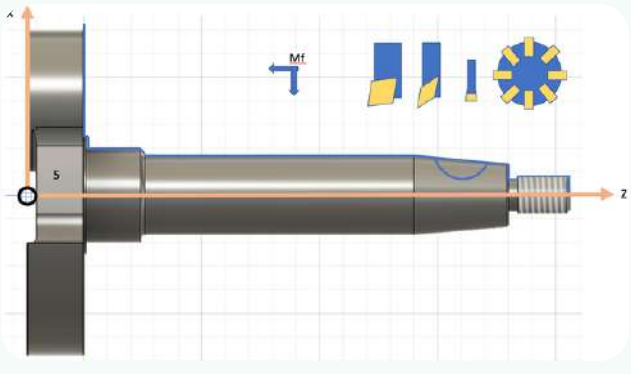
PHASE N°20 — Element : Vilebrequin — Machine : centre d'usinage millturn 5 axes



Opération :	Outil :	Vc	f	ap
dressage ébauche	outil carbure εr=85	100	0,3	1
dressage finition	outil carbure εr=35	150	0,1	0,5
ébauche du contour	fraise 2 taille diamètre 12	100	0,15	2
finition du contour	fraise 2 taille diamètre 12	150	0,1	3
ébauche de la poche	fraise torique rayon 1,5	100	0,15	1
finition de la poche	fraise torique rayon 1,5	150	0,1	0,5
pointage	foret diamètre 10mm	20	0,05	/
percage	foret diamètre 14mm	20	0,1	/

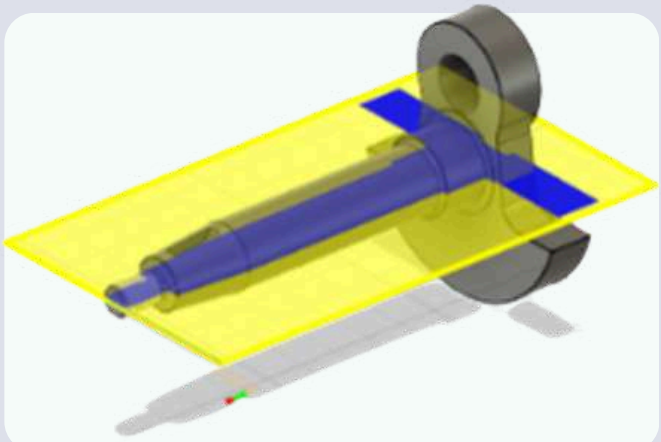
01 Machining process plan

Phases 10 & 20: operations, tools and cutting data (Vc, f, ap)



02 Workpiece clamping

Datum, clamping and tool selection for phase 10



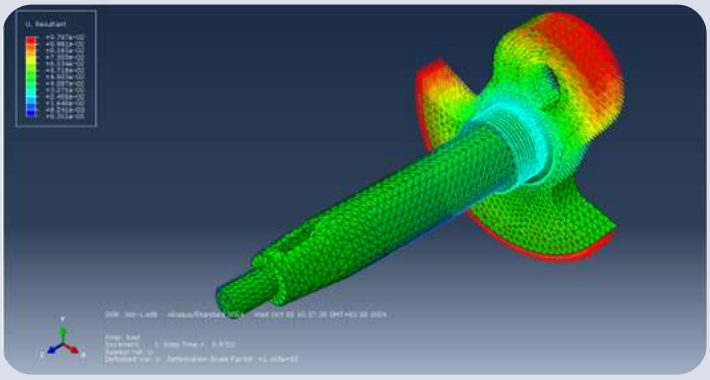
03 Machining simulation

Toolpath programming and verification in Fusion 360

04 / Fatigue & Fracture Mechanics

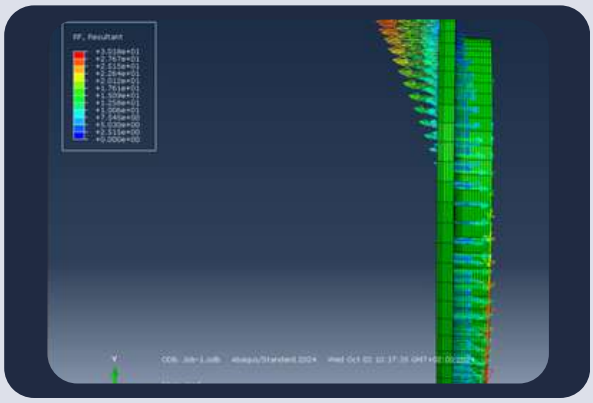
Finite Element Analysis with Abaqus

Simulated cyclic loading on a crankshaft to assess its fatigue life with the Crossland criterion, and modelled a cracked specimen to determine its fracture load.

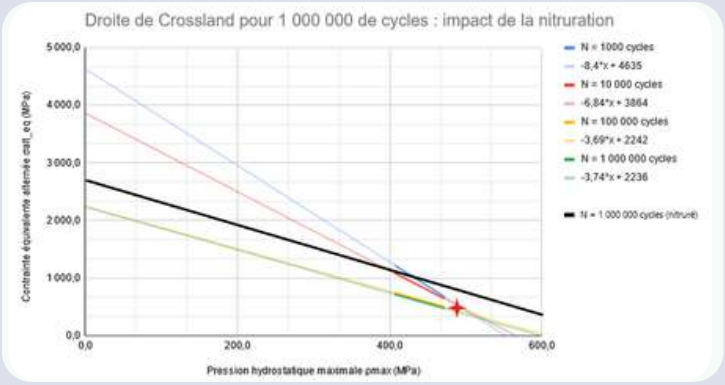


Crankshaft under cyclic load — Abaqus

A • Cyclic fatigue of a crankshaft

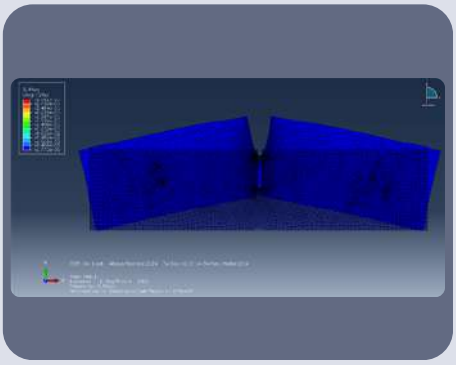


01 Loading & reactions
Boundary conditions and reaction forces on the shaft

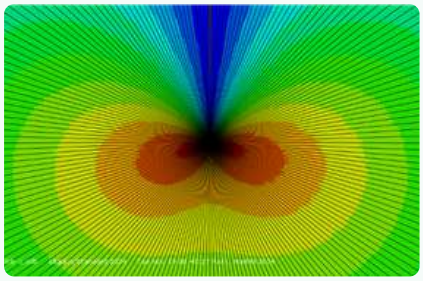


02 Crossland fatigue criterion
Endurance at 10^6 cycles: impact of nitriding

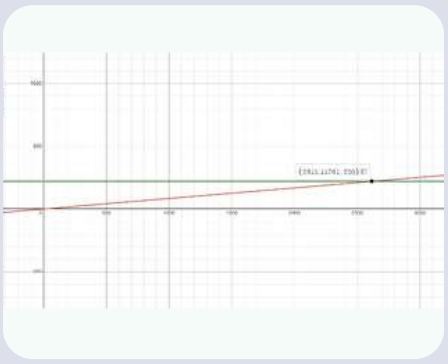
B • Fracture mechanics



01 Cracked specimen
Notched specimen model under load



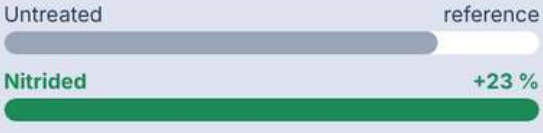
02 Crack-tip stress field
Stress concentration at the crack tip



03 Fracture load
Determined from the extracted results

C • Results

+23%
fatigue strength at 10^6 cycles



Nitriding raises the Crossland endurance limit of the crankshaft.

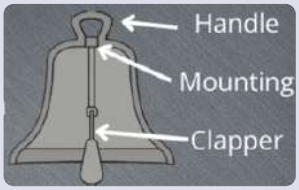
2615 N
fracture load

Critical load at rupture
Load at which the crack propagates, extracted from the Abaqus results.

05 / Foundry — Bell Components

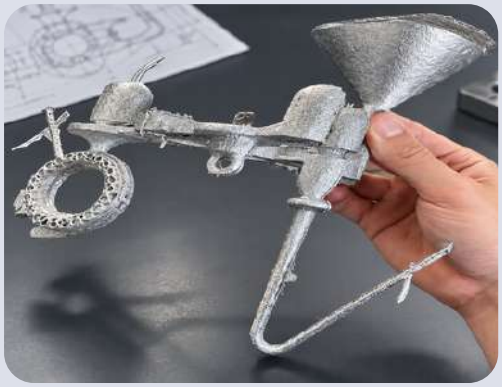
Casting design and filling & solidification simulation

Designed the casting cluster and mould for three bell components in CATIA V5, simulated mould filling and shrinkage in QuickCast, then cast the parts.



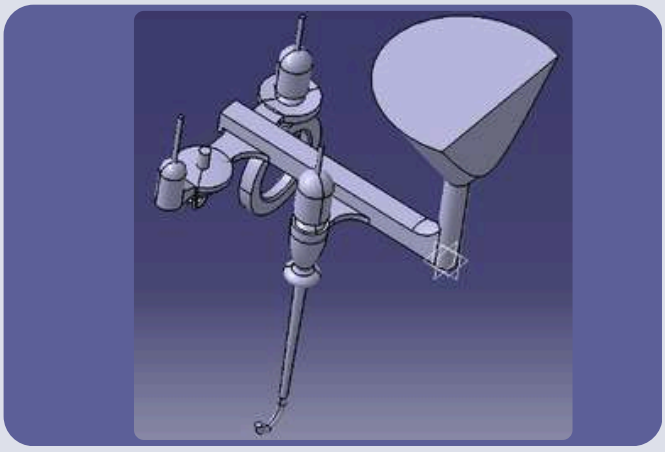
Cast parts

Handle, mounting and clapper, cast together in one cluster



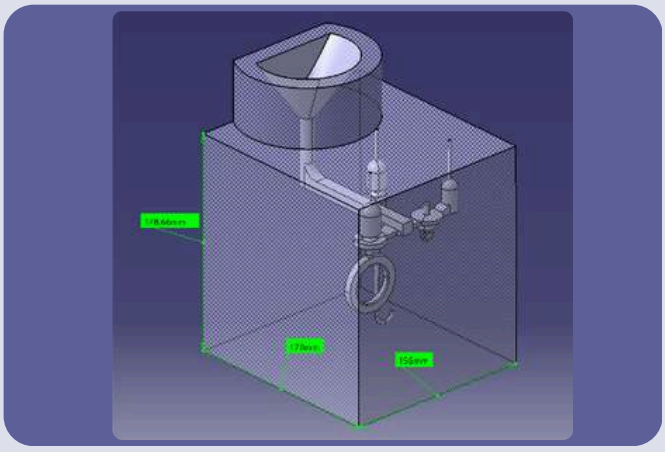
Cast cluster after shake-out

Casting process



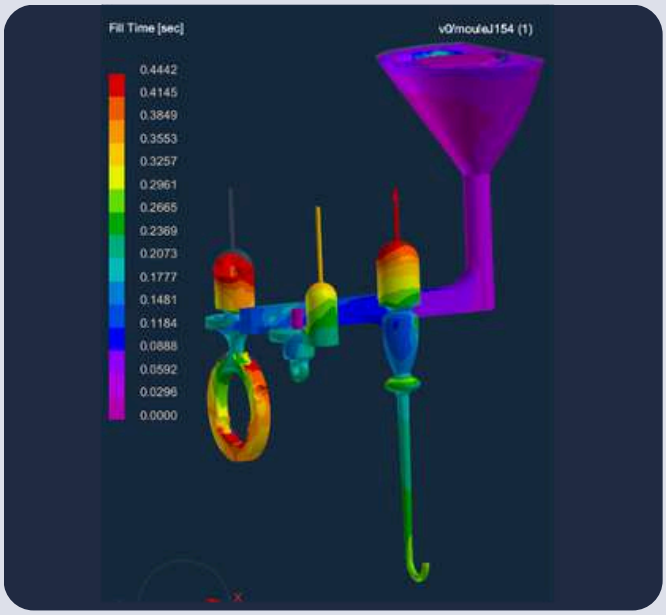
01 Casting cluster

Parts, runners, risers and pouring cup — CATIA V5



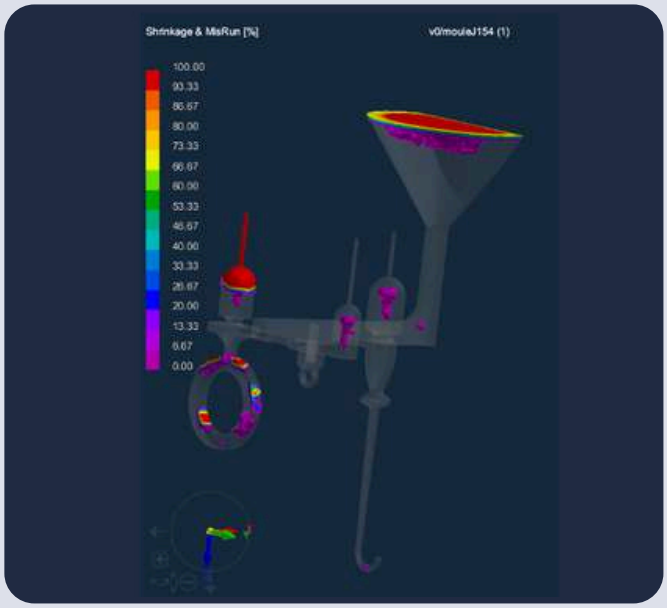
02 Mould design

Cluster positioned in the mould with dimensions



03 Filling simulation

Fill time of each part — QuickCast



04 Shrinkage & misrun check

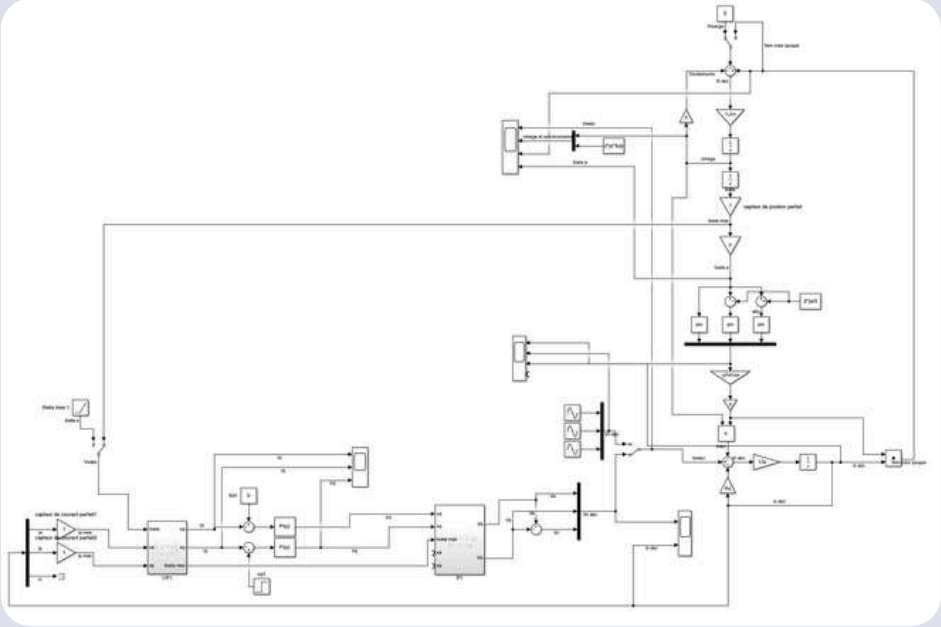
Risk areas identified before pouring

06 / Modelling & Control Engineering

From electrical machine modelling to real-time position control

Modelled a synchronous machine from its equations in MATLAB Simulink, then built a scaled wind-turbine nacelle and closed its position loop on a Raspberry Pi Pico, comparing several control strategies.

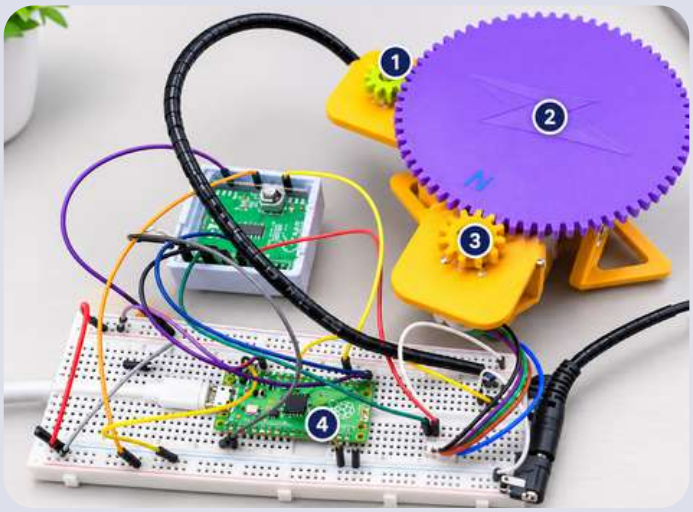
A • Synchronous machine model



Equation-based model in MATLAB Simulink

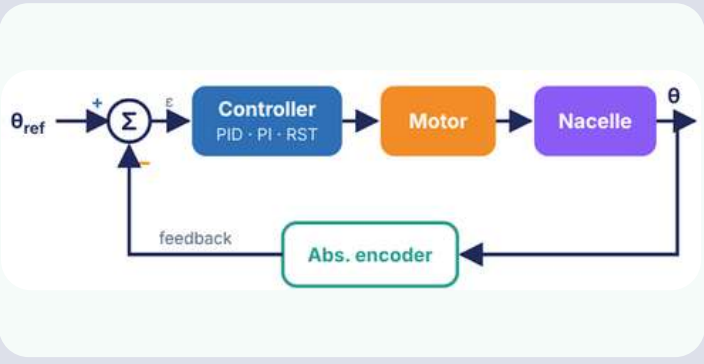
Implemented the machine's governing equations as a block diagram to simulate its dynamic behaviour.

B • Wind turbine nacelle position control



01 Physical prototype

1. Absolute encoder
2. Nacelle (3D-printed)
3. Electric motor
4. Raspberry Pi Pico



02 Closed-loop architecture

Measured angle fed back and compared to the setpoint at each cycle

03 Controllers compared

PID	PI	RST
proportional–integral–derivative	proportional–integral	polynomial controller